

SOCIAL MEDIA INFLUENCE ON TEEN LIFESTYLE & WELL-BEING

Power BI Dashboard Project Report

1. Project Information

Project Title: Social Media Influence on Teen Lifestyle & Well-Being

Tools Used: Excel, Power Query, Power BI, DAX

Dataset Source: [Kaggle Teen Social Media Usage & Mental Health](#)

2. Project Overview

This project analyzes social media usage patterns and their relationship with wellness, lifestyle behaviors, and academic outcomes among adolescents.

3. Project Objectives

- Analyze social media usage patterns
- Evaluate mental well-being
- Assess lifestyle impacts
- Examine academic outcomes

4. Dataset Overview

Metrics	Value
Total teens	2500
Age Range	13-19
Total Platforms	A. Instagram B. Tik Tok C. Both D. Other
Key Attributes	1. Age 2. Gender 3. Daily Social Media Hours 4. Platform Usage 5. Sleep Hours

	6. Screen Time Before Sleep 7. Academic Performance 8. Physical Activity 9. Social Interaction Level 10. Stress Level 11. Anxiety Level 12. Depression Label
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5. Data pre-processing using Excel

Process	Step
Checked duplicates	Home → Keep Rows → Keep Duplicates, no duplicate entries were found
Handled missing values	View- Column Quality- Missing value analysis was performed. No null or blank values were detected
Standardized formats	a) Converted the First letters of column names to Capital letters b) In "Gender" column, Converted 'male' and 'female' to 'Male' and 'Female' c) In "Social_interaction_level" and "Depression_risk" columns converted 'low', 'medium' and 'high' to 'Low', 'Medium' and 'High' d) Select "Academic_performance"- Transform- Rounding- To '1' decimal place e) Applied Clean and Trim
Corrected data types	Home → Data type, Checked and corrected all the columns data type

Calculated Columns Created

Column Name	Conditions
Mental Strain Index (MSI)	Add column → Custom column → (Stress level + Anxiety level)/2
Mental Strain Status	Add column → Conditional column → "Mental Strain Index (MSI) "Less than or equal to 5- "Healthy", Less than or equal to 10- "Moderate Concern", Else- "High Risk"
Age Group	Add column → Conditional column → "Age" Less than or equal to 14- "Early Teen", Less than or equal to 18- "Mid Teen", Else- "Late Teen"
Sleep Quality	Add column → Conditional column → "Sleep hrs" Less than or equal to 6- "Poor Sleep", Less than or equal to 8.9- "Healthy Sleep", Else- "Over Sleep"
Academic Status	Add column → Conditional column → "Academic Performance" Less than or equal to 2.49- "Below Average", Less than or equal to 2.99- "Average", Less than or equal to 3.49- "Good", Else- "Excellent"
Social Media Usage Status	Add column → Conditional column → "Screen time before Sleep" Less than or equal to 1- "Low Usage", Less than 2- "Medium Usage", Else- "High Usage"
Physical Activity Status	Add column → Conditional column → "Physical Activity" Less than 1- "Poor Activity", Less than or equal to 1.5- "Moderate Activity", Else- "High Activity"

Pivot Tables Created

- A) Platform wise Average Stress
- B) Gender wise Depression Risk
- C) Age group wise Average Social media hours
- D) Academic Performance by Average Screen time before sleep
- E) Mental Strain Status by Physical Activity

6. Dashboard Architecture

Page 1 – Overview & Navigation

Provides project introduction, KPIs, and dashboard navigation.

Page 2 – Social Media Usage Analytics

Analyzes platform usage, age groups, gender distribution, and usage intensity.

Page 3 – Mental Health Impact

Evaluates stress, anxiety, depression risk, and Mental Strain Index.

Page 4 – Lifestyle & Physical Health Impact

Explores sleep quality, physical activity, and social interaction.

Page 5 – Academic Influence

Examines academic performance in relation to digital habits and wellness indicators.

Page 6 – Decomposition Tree Analysis

Identifies the key pathways leading to high mental strain among teenagers.

Page 7 – Conclusion & Recommendations

Summarizes findings and suggests actionable recommendations.

➤ Features & Tools Used

Power BI Features

- ☐ Interactive multi-page dashboard design
- ☐ Cards for key performance indicators
- ☐ Slicers for dynamic filtering and data exploration
- ☐ Page Navigation Buttons for seamless report navigation
- ☐ Custom backgrounds, icons, and visual themes
- ☐ Tooltips and drill-down capabilities for detailed analysis

Advanced Visualizations

- ☐ **Gauge Chart** – Mental Strain Index (MSI) performance tracking
- ☐ **Waterfall Chart** – Contribution of stress, anxiety, sleep, and activity to mental strain
- ☐ **Ribbon Chart** – Ranking changes across categories
- ☐ **Funnel Chart** – Teen Well-Being Risk Escalation analysis

- ❑ **Treemap** – Platform and social media usage distribution
- ❑ **Scatter Chart** – Relationship between stress and media usage
- ❑ **Line & Area Charts** – Trend analysis
- ❑ **Matrix** – multi-dimensional comparison of key metrics
- ❑ **Decomposition Tree** – Root-cause analysis of high mental strain

7. DAX Measures

A) KPI Measures

Measure	Formula
Total Teens	COUNTROWS(Teen_Mental_Health_Dataset)
Total Male	CALCULATE ([Total Teens], Teen_Mental_Health_Dataset [Gender] = "Male")
Total Female	CALCULATE ([Total Teens], Teen_Mental_Health_Dataset [Gender] = "Female")
Platforms Used	DISTINCTCOUNT (Teen_Mental_Health_Dataset [Platform_usage])

B) Wellness Measures

Measure	Formula
Avg Stress	AVERAGE(Teen_Mental_Health_Dataset[Stress_level])
Avg Anxiety	AVERAGE(Teen_Mental_Health_Dataset[Anxiety_level])
Avg MSI	AVERAGE(Teen_Mental_Health_Dataset[Mental Strain Index(MSI)])
High MSI	CALCULATE([Total Teens], Teen_Mental_Health_Dataset[Mental Strain Status] = "High Risk")
High Risk	CALCULATE([Total Teens], Teen_Mental_Health_Dataset[Mental Strain Status] = "High Risk")
High Risk %	DIVIDE([High Risk], [Total Teens])

C) Lifestyle Measures

Measure	Formula
Avg Sleep	AVERAGE(Teen_Mental_Health_Dataset[Sleep_hours])
Avg Physical Activity	AVERAGE(Teen_Mental_Health_Dataset[Physical_activity])

D) Usage Measures

Measure	Formula
Avg Social Media Hours	AVERAGE(Teen_Mental_Health_Dataset[Daily_social_media_hours])
High Media Users	CALCULATE([Total Teens], Teen_Mental_Health_Dataset[Social Media Usage Status] = "High Usage")

E) Gauge Measures

Measure	Formula
Min-MSI	MIN(Teen_Mental_Health_Dataset[Mental Strain Index(MSI)])
Max-MSI	MAX(Teen_Mental_Health_Dataset[Mental Strain Index(MSI)])
Target MSI	5
Avg MSI	AVERAGE(Teen_Mental_Health_Dataset[Mental Strain Index(MSI)])

F) Waterfall Measures

Measure	Formula
Waterfall Value	SWITCH(SELECTEDVALUE(WaterfallFactors[Factor]), "Stress",[Avg Stress], "Anxiety",[Avg Anxiety], "Sleep",- [Avg Sleep], "Physical Activity",- [Avg Physical Activity], "Final MSI",[Avg MSI], BLANK())

G) Academic Measures

Measure	Formula
Avg Academic Performance	AVERAGE(Teen_Mental_Health_Dataset[Academic_performance])

Power BI Report (PDF Export): [Click here to view Power BI Dashboard \(PDF\)](#)

10. Key Findings

Social Media Usage

- Average daily social media usage is approximately 4.46 hours.
- Usage patterns vary across age groups and genders.

Mental Well-Being

- Higher usage levels are associated with higher Mental Strain Index scores.
- Stress and anxiety are major contributors to mental strain.

Lifestyle Impact

- Poor sleep quality is linked to higher mental strain.
- Lower physical activity is associated with higher risk levels.

Academic Impact

- Better sleep and lower mental strain generally correspond to stronger academic outcomes.

Decomposition Tree Insights

- High social media usage, poor sleep, low physical activity, and high depression risk form the most significant pathway leading to high mental strain.
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11. Conclusion

The analysis indicates that social media usage has a measurable influence on adolescent well-being, lifestyle habits, and academic performance. High usage levels combined with poor sleep and low physical activity are associated with increased mental strain among teenagers.

12. Recommendations

Promote Responsible Social Media Usage

Encourage balanced and mindful digital engagement.

Improve Sleep Hygiene

Reduce screen exposure before bedtime and maintain consistent sleep schedules.

Encourage Physical Activity

Promote regular exercise and outdoor activities.

Strengthen Social Interaction and Support Academic Well-Being

Support healthy offline social engagement and encourage effective time management and healthy study habits